Cluster Sampling with Equal Probabilities

STA304: Surveys, Sampling, and Observational Data

Emily Somerset, Thursday May 28, 2026

Learning Outcomes

- [ ] Relate the efficiency of a cluster sample to the amount of within-cluster and between-cluster variation
- [ ] Interpret the within-cluster and between-cluster components of variation using an ANOVA decomposition.
- [ ] Explain why cluster sampling is often less precise than a simple random sample of the same size.
- [ ] Use information about within-cluster variation, between-cluster variation, and the ICC to make recommendations about cluster sampling designs.
- [ ] Balance precision and cost considerations when choosing the number of clusters and the number of units sampled within each cluster.

Last class

$\bar{y}_i < \bar{y}_j$

- We talked about one-stage and two-stage cluster sampling designs.
- We talked about how these cluster sampling designs produce estimators that are in general less efficient than those under an SRS.
    - e.g. MSE$(\hat{t}_{\text{cluster}})$ $>$ MSE$(\hat{t}_{\text{SRS}})$ for a nM observations

      write the $V(\hat{t}_{cluster})$ as function of a measure of homogeneity
    - In one-stage cluster design: These nM observations come from n randomly selected clusters and M not randomly selected units in each sampled cluster (census).
    - Units in the same cluster are generally homogeneous in some important way, often unmeasured way, related to the outcome of interest y
    - So every additional observation within a cluster doesn't offer too much "new" information.
    - This is the intuition behind what is happening.

Last class

- To understand this intuition a bit deeper we will look at an ANOVA table.
- This summarizes data by partitioning total data variability into specific sources: Within-group variability and between-group variability.

5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- This is an ANOVA table
- It decomposes the total variance

TABLE 5.1
Population ANOVA Table—Cluster Sampling

Source	df	Sum of Squares	Mean Square
Between psus	$N-1$	$SSB = \sum_{i=1}^{N}\sum_{j=1}^{M}(\bar{y}_{iU} - \bar{y}_U)^2$	MSB
Within psus	$N(M-1)$	$SSW = \sum_{i=1}^{N}\sum_{j=1}^{M}(y_{ij} - \bar{y}_{iU})^2$	MSW
Total, about $\bar{y}_U$	$NM-1$	$SSTO = \sum_{i=1}^{N}\sum_{j=1}^{M}(y_{ij} - \bar{y}_U)^2$	S^2

5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

$\bar{y}_{2\cdot}$ $\bar{y}_{1\cdot}$ $\bar{y}_{\cdot\cdot}$

- This is an ANOVA table
- It decomposes the total variance
- Into the sum of the within-group and between-group variance.

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Population ANOVA Table—Cluster Sampling

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5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- **Stop and think**: Take a look at each of these formulas and explain in your own words what it measures.

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5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- The degree of information measures how many **independent** pieces of information you have.

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5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- Example:

$$SSB = \sum_{i=1}^{N}\sum_{j=1}^{M}(\bar{y}_{i\mathcal{U}} - \bar{y}_{\mathcal{U}})^2 = M\sum_{i=1}^{N}(\bar{y}_{iU} - \bar{y}_U)^2$$

- So first we have computed N pieces of information $\bar{y}_{1U}, \bar{y}_{2U}, \ldots, \bar{y}_{NU}$
- $d_i = \bar{y}_{iU} - \bar{y}_U$

1) $\sum_{i=1}^{N} d_i = 0$ constraint.

$$\sum_{i=1}^{N} d_i = \sum_{i=1}^{N}(\bar{y}_{iU} - \bar{y}_U) = \sum_{i=1}^{N}\bar{y}_{iU} - N\bar{y}_U = N\bar{y}_U - N\bar{y}_U = 0$$

$$\bar{y}_U = \frac{1}{N}\sum_{i=1}^{N}\bar{y}_{iU}$$

$$SSB = M\sum_{i=1}^{N} d_i^2$$

SSB is computed with $N-1$ pieces of independent information

5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- The mean square column is the sum of squares column divided by the degrees of freedom.

TABLE 5.1
Population ANOVA Table—Cluster Sampling

Source	df	Sum of Squares	Mean Square
Between psus	$N-1$	$\text{SSB} = \sum_{i=1}^{N}\sum_{j=1}^{M}(\bar{y}_{iU}-\bar{y}_U)^2$	MSB $= \frac{SSB}{N-1}$
Within psus	$N(M-1)$	$\text{SSW} = \sum_{i=1}^{N}\sum_{j=1}^{M}(y_{ij}-\bar{y}_{iU})^2$	MSW $= \frac{SSW}{N(M-1)}$
Total, about $\bar{y}_U$	$NM-1$	$\text{SSTO} = \sum_{i=1}^{N}\sum_{j=1}^{M}(y_{ij}-\bar{y}_U)^2$	$S^2 = \frac{SSTO}{NM-1}$

5.4 Comparing Clusters with an SRS

Connection to the total estimator

- The variance of the total estimator was:
- $$V(\hat{t}) = N^2 \frac{S_t^2}{n}\left(1 - \frac{n}{N}\right)$$
- And depends on S_t^2

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Total, about $\bar{y}_U$	$NM-1$	$SSTO = \sum_{i=1}^{N}\sum_{j=1}^{M}(y_{ij} - \bar{y}_U)^2$	S^2

$$S_t^2 = \frac{1}{N-1}\sum_{i=1}^{N}(t_i - t/N)^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(M\bar{y}_{iU} - M \times N\bar{y}_U/N\right)^2$$

$$= \frac{1}{N-1}\sum_{i=1}^{N}(M\bar{y}_{iU} - M\bar{y}_U) = M \times \frac{SSB}{N-1} = \boxed{M \times MSB}$$

5.4 Comparing Clusters with an SRS

Connection to the total estimator

$t_1 = M\bar{y}_{1U}$ $t_2 = M\bar{y}_{2U}$

- The variance of the total estimator was: $S_t^2 = M \times MSB$

- $$V(\hat{t}) = N^2 \frac{S_t^2}{n}\left(1 - \frac{n}{N}\right)$$

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$\hat{t} = N\bar{t}_S$

- And so in terms of MSB:

- $V(\hat{t}) = N^2 \times \frac{M \times MSB}{n}\left(1 - \frac{n}{N}\right) = \frac{N^2 M}{n}\left(1 - \frac{n}{N}\right) \times MSB$

- as the between cluster variance increases so does the sampling variance of $\hat{t}$

5.4 Comparing Clusters with an SRS

Connection to the total estimator

- $$V(\hat{t}) = \frac{N^2 M}{n}\left(1 - \frac{n}{N}\right) \times \text{MSB}$$
- Note on this intuitively:

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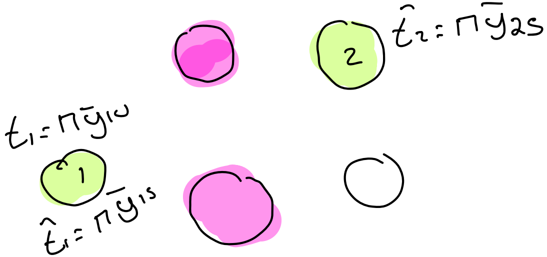
$\hat{t} = N\bar{t}_s$



$\hat{t} = N\left(\frac{1}{2}[\hat{t}_1 + \hat{t}_2]\right)$

since t_1, t_2, t_3, etc... are so different then $\hat{t}$ will really depend on which cluster we sample and it will really vary alot from sample to sample.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- 1-stage cluster nM observations:
- $V(\hat{t}_{\text{cluster}}) = \dfrac{N^2 M}{n}\left(1 - \dfrac{n}{N}\right) \times \text{MSB}$
- SRS with nM observations:
- $V(\hat{t}_{\text{SRS}}) = (NM)^2 \dfrac{S^2}{nM}\left(1 - \dfrac{nM}{NM}\right)$

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5.4 Comparing Clusters with an SRS

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- 1-stage cluster nM observations:

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- SRS with nM observations:

- $$V(\hat{t}_{\text{SRS}}) = (NM)^2 \frac{S^2}{nM}\left(1 - \frac{nM}{NM}\right)$$

- Relative efficiency:

$$= MSB / S^2$$

- $$\frac{V(\hat{t}_{\text{cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{\frac{N^2 M}{n}\left(1 - \cancel{\frac{n}{N}}\right) \times MSB}{\frac{(NM)^2}{nM}\left(1 - \cancel{\frac{nM}{NM}}\right) \times S^2}$$

$$= \frac{\frac{N^2 M}{n}}{\left(\frac{N^2 M^2}{nM}\right)} \times \frac{MSB}{S^2} = \boxed{\frac{MSB}{S^2}}$$

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

$\rightsquigarrow$ MSB > S^2 then $\hat{t}_{\text{cluster}}$ is less efficient than S^2

- Let's relate $\dfrac{V(\hat{t}_{\text{cluster}})}{V(\hat{t}_{\text{SRS}})} = \dfrac{\text{MSB}}{S^2}$ to the intraclass coefficient ICC.

- Recall from last class: The ICC provides a measure for **homogeneity within clusters**

- It's ~~denied~~ defined as the Pearson correlation coefficient for the $NM(M-1)$ pairs of $\underline{(y_{ij}, y_{ik})}$ observations for $i = 1,\ldots,N, j \neq k$

$$\text{ICC} = \frac{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})(y_{ik} - \bar{y}_{\mathcal{U}})}{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})^2}$$

Measures the correlation between all pairs from a same cluster across the population.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- It can be shown that:

$$\text{ICC} = \frac{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})(y_{ik} - \bar{y}_{\mathcal{U}})}{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})^2} = 1 - \frac{M}{M-1} \frac{\text{SSW}}{SSTO}$$

$$0 \leq \frac{\text{SSW}}{SSTO} \leq 1 \qquad -\frac{1}{M-1} \leq \text{ICC} \leq 1$$

SSTO = SSW + SSB

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5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

$$\text{ICC} = \frac{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})(y_{ik} - \bar{y}_{\mathcal{U}})}{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})^2} = 1 - \frac{M}{M-1} \frac{\text{SSW}}{SSTO} \quad - $$

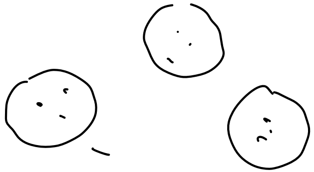
result

- As within-cluster variation increases, clusters are more homogeneous, and ICC approaches 1
- As within cluster variation decreases, clusters are more heterogeneous, and ICC approaches $-\frac{1}{n-1}$

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- ICC < 0 when $\dfrac{\text{SSW}}{SSTO} > \dfrac{M-1}{M}$



- If the Pearson correlation is negative, within each cluster, high and low values are systematically paired together.
- A randomly assigned group of people into each cluster would just reflect population variance.
- But if clustering actively mixes opposites together, the within spread exceeds what you would get by chance -> **more heterogeneous than the population.**

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

Example: consider two populations each with 3 psu and 3 ssu in each psu:

	Population A			Population B		
psu 1	10	20	30	9	10	11
psu 2	11	20	32	17	20	20
psu 3	9	17	31	31	32	30

	Population A		Population B	
	$\bar{y}_{iU}$	S_i^2	$\bar{y}_{iU}$	S_i^2
psu 1	20	100	10	1
psu 2	21	111	19	3
psu 3	19	124	31	1

- Both populations have the same elements (numbers) just organized differently
- So they share $\bar{y}_{\mathcal{U}} = 20$ and $S^2 = 84.5$

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

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	Population A		Population B	
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psu 1	20	100	10	1
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psu 3	19	124	31	1

- In Population A, the means of each cluster are similar and there's lots of within-cluster variability (S_i^2)
- In Population B, there's less within-cluster variability (S_i^2) but more variation between the cluster means.

5.4 Comparing Clusters with an SRS

$$1 - \frac{M}{M-1}\frac{\text{SSW}}{SSTO}$$

Revisiting comparing precision with an SRS of the same size

Example: consider two populations each with 3 psu and 3 ssu in each psu:

ANOVA Table for Population A:

Source	df	SS	MS
Between psus	2	6	3
Within psus	6	670	111.67
Total, about mean	8	676	84.5

ANOVA Table for Population B:

Source	df	SS	MS
Between psus	2	666	333
Within psus	6	10	1.67
Total, about mean	8	676	84.5

$$\text{ICC} = 1 - \left(\frac{3}{2}\right)\frac{670}{676} = -0.4867$$

$$\text{ICC} = 1 - \left(\frac{3}{2}\right)\frac{10}{676} = 0.9778$$

- We can see that the ICC in population A is negative and positive in Population B.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

	Population A			Population B		
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psu 2	11	20	32	17	20	20
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	Population A		Population B	
	$\bar{y}_{iU}$	S_i^2	$\bar{y}_{iU}$	S_i^2
psu 1	20	100	10	1
psu 2	21	111	19	3
psu 3	19	124	31	1

- In population A, every cluster has a small, medium, and large value.
- So if you pick two elements from the same cluster, you're guaranteed to get one low one high (negatively correlated).
- This is more dispersed than what you'd expect just drawing from the entire population.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

$$\frac{V(\hat{t}_{\text{cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{NM-1}{M(N-1)}[1+(M-1)\text{ICC}]$$

← measures relative efficiency of $\hat{t}_{cluster}$ to $\hat{t}_{SRS}$

result

Summary:

- Only when ICC is negative, would cluster estimators be more efficient than SRS estimators.
- If ICC is positive, PSUs in the same cluster tend to be similar, and cluster sampling is less efficient than simple random sampling of elements.

5.5 Designing a Cluster Sample

n, M, m

n

Design Questions

1. What overall precision is needed?
2. What size should the psus be? (AKA how big should M be?)
3. How many ssus should be sampled in each psu selected for the sample?
4. How many psus should be sampled?

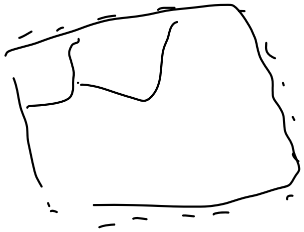
Q1 is faced in any survey design. Often a conversation with experts.

Forward sortation area
FSA

5.5 Designing a Cluster Sample

Size of the psus

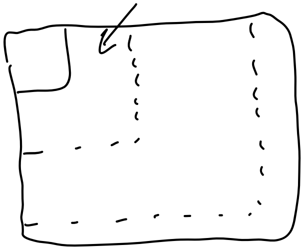
- Often psus are naturally forming and this decision is made for us.
  - E.g. classrooms, schools, farms, egg cartons
- This means our decision on the size of M would be made.
- Can be more difficult like splitting up a city into geographical areas.
- How large should each psu be? 1km^2, 10km^2, 20km^2 ?



5.5 Designing a Cluster Sample

Size of the psus

- General principle in area surveys is that the larger the psu the better because the more variability you are likely to see in a psu.
- However if the psu size is too large, you might lose the cost savings of cluster sampling.



5.5 Designing a Cluster Sample

Choosing subsampling sizes

- Suppose we have chosen a psu size
- Let $M_i = M$ and $m_i = m$ for all psus (simplest case)
- Then:

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

5.5 Designing a Cluster Sample

Choosing subsampling sizes

- Suppose we have chosen a psu size
- Let $M_i = M$ and $m_i = m$ for all psus.
- Then:

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right)\frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right)\frac{\text{MSW}}{nm}$$

← variance

If MSW = 0 (no within-variation), you may as well set $m = 1$. Adding more subsamples doesn't ~~decrease~~ increase precision of the estimator.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right)\frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right)\frac{\text{MSW}}{nm}$$

Often you'll combine this variance formula with a cost function:

Total cost = C = $n(c_1 + c_2 m)$ $= \underbrace{n c_1}_{\text{cost per cluster}} + \underbrace{c_2}_{\text{cost per subsample}} \overbrace{n \times m}$

Example: $c_1 = \$30$ = cost to visit a neighbourhood (cluster)

$c_2 = \$10$ = cost to visit each house (ssu) in neighbourhood.

eg $n=1$, $m=2$

$C = \$30 + \$10 \times 2 = \$50$

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right)\frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right)\frac{\text{MSW}}{nm}$$

Often you'll combine this variance formula with a cost function:

Total cost = C = $n(c_1 + c_2 m)$

Example: $c_1 = 40$ minutes per dorm room

$c_2 = 10$ minutes per student.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right)\frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right)\frac{\text{MSW}}{nm}$$

Often you'll combine this variance formula with a cost function:

Total cost = C = $n(c_1 + c_2 m)$

Now you have an optimization problem! Minimize V constrained to this cost function, for a fixed total cost C

5.5 Designing a Cluster Sample

Choosing subsampling sizes

The textbook actually has an exercise to solve this optimization problem. The answer is:

$$n_{\text{opt}} = \frac{C}{c_1 + c_2 m_{\text{opt}}}$$

$$m_{\text{opt}} = \sqrt{\frac{c_1 M(N-1)(1-R_a^2)}{c_2(NM-1)R_a^2}}$$

population quantity

Here $R_\alpha^2 = 1 - \frac{\text{MSW}}{S^2}$

Another measure of within-group homogeneity (adjusted R^2)

If you didn't have this population quantity you could estimate it using past survey data or other observational data.

$$\hat{R}_\alpha = 1 - \frac{\hat{\text{MSW}}}{\hat{S}^2}$$ (from ANOVA table)

5.5 Designing a Cluster Sample

Choosing subsampling sizes

at fixed cost, $\widehat{MSB}$, $\widehat{MSW}$ at fixed M

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right)\frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right)\frac{\text{MSW}}{nm}$$

Minimize this variance formula with respect to a cost function:

Total cost = C = $n(c_1 + c_2 m)$

Or you can plot the variance versus the cost for a variety of subsamples m and n.

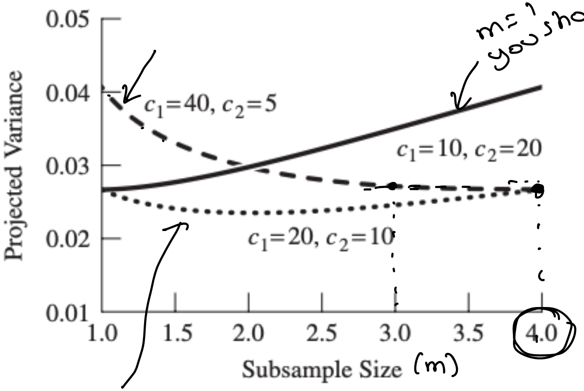
e.g dorm room
M = 4

5.5 Designing a Cluster Sample

Choosing subsampling sizes

Or you can plot the variance versus the cost for a variety of subsamples m.

m = 1
you should only interview one person per dorm



flat
min at approx ≈ 2 subsamples,
but any subsample is acceptable
here.

- Here $\hat{R}_\alpha$ was set based on past data
- C was fixed.
- c_1, c_2, m were varied and plotted against variance

eg. dorm room, If it takes 40 minutes per suite
and 5 min per person, one-stage sampling
should be used.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

- Summary:
  - For design purposes we only need an estimate of ICC or R_α^2 or MSW and MSB

adjusted R^2

We obtain the following ANOVA table for the coots data in Example 5.7.

Source	DF	Sum of Squares	Mean Square	F Value
Model	183	257.4175336	1.4066532	237.44
Error	184	1.0900782	0.0059243	
Corrected Total	367	258.5076118		

R-Square	C.V.	Root MSE	VOLUME Mean
0.995783	3.298616	0.076970	2.333394

High degree of homogeneity within groups ($R_\alpha^2 \approx 0.996$)

you would set $\hat{R}_\alpha^2 = 0.996$ when designing a cluster sample

5.5 Designing a Cluster Sample

Choosing the sample size (number of psus)

- Once you determine M and m and you have a desired precision, you can determine n usual methods.
- That is, construct the theoretical half width of a confidence interval and set it to the desired precision and solve for n!
- You can use past data to estimate parameters that may be needed (e.g. s_i^2, s_t^2).

$$\hat{V}(\hat{t}) \le \left(\frac{\text{pre}}{z_{\alpha/2}}\right)^2$$

$$\hat{V}(\hat{t}) = N^2\left(1 - \frac{n}{N}\right)\frac{s_t^2}{n} + \frac{N}{n}\sum_{i \in S}\left(1 - \frac{m_i}{M_i}\right)M_i^2\frac{s_i^2}{m_i}$$

$$z_{\alpha/2}\sqrt{\hat{V}(\hat{t})} = \text{Desired Precision}$$

5.5 Designing a Cluster Sample

Summary

- All these choices of M, n, and m might need to be reiterated until you find a combination that works.

- Increasing M to increase heterogeneity.

- And then figuring out an n and an m that will give a low variance with a desired level of precision.

Learning activity 8 will briefly touch on these ideas.

Progress

- ☑ Relate the efficiency of a cluster sample to the amount of within-cluster and between-cluster variation
- ☑ Interpret the within-cluster and between-cluster components of variation using an ANOVA decomposition.
- ☑ Explain why cluster sampling is often less precise than a simple random sample of the same size.
- ☑ Use information about within-cluster variation, between-cluster variation, and the ICC to make recommendations about cluster sampling designs.
- ☑ Balance precision and cost considerations when choosing the number of clusters and the number of units sampled within each cluster.